

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech.

Branch : Civil Engineering/Civil and Infra

Semester :V

Subject Code & Name:

Design of Steel Structures BTCVC501/BTCIC501

Max Marks: 60

Date: 05/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.
7. Use of IS 800-1984, 2007, IS-875 is permitted

		CO	Marks
Q.1 Objective type questions. (Compulsory Question)			12
1	What is one primary advantage of steel structures?	CO1	1
	a. Low durability		
	b. Ease of construction and fabrication		
	c. Limited availability of materials		
	d. Poor fire resistance		
2	What is the main function of bolted connections in steel structures?	CO1	1
	a. To provide aesthetic appeal		
	b. To distribute thermal loads uniformly		
	c. To join steel members securely		
	d. To increase the elasticity of the structure		
3	Which type of connection is considered more rigid and reliable in steel structures?	CO2	1
	a. Riveted		
	b. Bolted		
	c. Welded		
	d. Threaded		
4	Which failure mode is common in bolted connections under excessive tensile force?	CO2	1
	a. Shear failure of the bolt		
	b. Bearing failure of the plate		
	c. Rupture of the bolt		
	d. Tearing of the connected plate		
5	Compression members are more prone to:	CO3	1
	a. Buckling failure		
	b. Tearing failure		
	c. Tensile failure		
	d. Shear failure		
6	Which of the following parameters is used to calculate the effective area of a tension member?	CO3	1
	a. Gross area and gauge length		
	b. Gross area and shear lag factor		
	c. Net area and modulus of elasticity		
	d. Radius of gyration and net area		

7	For a compression member with both ends fixed, the effective length factor (K) is:	CO3	1
	a. 1.0 b. 0.65 c. 0.5 d. 2.0		
8	In a laterally unsupported beam, failure occurs primarily due to:	CO3	1
	a. Shear stress b. Local buckling of the web c. Lateral-torsional buckling d. Bearing stress		
9	Web stiffeners in a beam are used to resist:	CO1	1
	a. Shear forces and prevent web buckling b. Bending moment and torsion c. Axial compression and lateral buckling d. Torsional deformation		
10	The most common force acting on a gantry girder is:	CO1	1
	a. Axial tension b. Vertical load and lateral thrust from crane operation c. Wind load from the roof d. Shear force from purlins		
11	The formation of a plastic hinge in a beam indicates:	CO4	1
	a. The member is undergoing elastic deformation. b. The member has reached its ultimate bending moment capacity. c. The structure is unstable and will collapse immediately. d. The beam is experiencing axial compression		
12	In Limit State Design, the partial safety factor for material strength is greater than 1 to account for:	CO4	1
	a. Load uncertainty b. Variability in material properties c. Deflection limits d. Dynamic effects		
Q.2	Solve the following.		12
A)	Define- 1)Pitch 2)Edge distance	CO2	6
	Determine the strength of rivet value of 20 mm dia. rivets connecting 12 mm. thick plate in single shear and in bearing. Permissible stress for rivet in shear and bearing are 80 N/mm ² and 250 N/mm ² and for plate in bearing is 250 N/mm ² .		
B)	1) Explain different types of loads acting on an industrial building 2) Explain the term Permissible stresses and state the permissible stresses in axial tension, axial compression, bending	CO1	6

- Q.3 Solve the following.** **12**
- A) Bottom tie of a 16 m roof truss is subjected to a tension of 52.5kN.Design the tie using angle section. CO3 6
- B) Explain with neat sketches web buckling & web crippling. CO3 6
- Q.4 Solve Any Two of the following.** **12**
- A) A truss of span 18 m. and of 4m spacing between trusses are used in the region where there is no snowfall. The pressure intensity due to wind is 1500 m².G.I. sheets are used for roofing. Calculate Dead load and live load per panel point. CO1 6
- B) Explain the role of purlins in a roof structure and their design. CO3 6
- C) Explain the different loads considered in the design of Gantry girder. CO3 6
- Q.5 Solve Any Two of the following.** **12**
- A) Design a single angle discontinuous strut to carry a load of 90 kN. The length of strut between the connections is 3m.Consider that angles are provided on opposite side of 12 mm gusset plate and assuming allowable compressive stress as 60 MPa design the strut. CO2 6
- B) Explain with neat sketches Lacing and batten systems in detail. CO2 6
- C) What are the different types of column bases? Explain with neat sketches. CO2 6
- Q.6 Solve Any Two of the following.** **12**
- A) Explain the process of hinge formation in a beam during plastic analysis. CO4 6
- B) Explain the major differences between the working stress method and limit state method as per IS 800-2007. CO4 6
- C) How steel structure design based on plastic analysis method differs from elastic design method. CO4 6

*** End ***